

# Introduction of the LAUREATE dataset: the Longitudinal multimodal student experience dataset for Affect and memory research

Matías Laporte  
matias.laporte@usi.ch

Università della Svizzera italiana  
Lugano, Ticino, Switzerland

Martin Gjoreski  
martin.gjoreski@usi.ch

Università della Svizzera italiana  
Lugano, Ticino, Switzerland

Enqi Fu  
enqi.fu@usi.ch

Università della Svizzera italiana  
Lugano, Ticino, Switzerland

Marc Langheinrich  
marc.langheinrich@usi.ch

Università della Svizzera italiana  
Lugano, Ticino, Switzerland

## ABSTRACT

Modern wearable devices enable the recording of physiological data from users in a continuous and unobtrusive manner. Using individual wearable devices to monitor students' psycho-physiological states could be helpful for personalised learning interventions to improve their understanding and memory recall of challenging topics. Working towards personalised interventions for students, we conducted a longitudinal study throughout a university semester with 44 participants (including two lecturers) in two courses. We recorded their physiological signals using a wristband device. We also administered pre-experiment, post-experiment and daily surveys, asking the students about their daily behaviour (e.g. study hours, smoking habits, physical activity, caffeine and food intake) and their perceived engagement, attention and emotions after class. Data was collected during the whole semester, totalling 26 sessions for each course, including their examinations (in-class quizzes, midterm and final exams). Hereby, we present the LAUREATE dataset, a Longitudinal multimodal student experience for Affect and memory dataset. The full dataset contains over 1400 hours of physiological data from 44 individuals, 3600 completed daily surveys from students, 70 hours of audiovisual recordings from the classes, and students' grades for quizzes, assignments and midterm and final exams, among others. To the best of our knowledge, this is the largest dataset for physiological data from the Empatica E4 device. We plan to make the dataset available for scientific purposes under a sharing agreement. Besides the initial intention of the dataset, i.e., to design post-class interventions to improve students' understanding and memory recall, the dataset is quite versatile and enables applications in many domains, including but not limited to the areas of affective computing, privacy, behaviour and learning performance modelling, context recognition, multimedia signal analysis, and the development of multi-modal machine learning algorithms.

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## CCS CONCEPTS

• **Human-centered computing** → *Collaborative and social computing; HCI design and evaluation methods.*

## KEYWORDS

Dataset; Students; Wearable Sensors; Memory Recall; Affect

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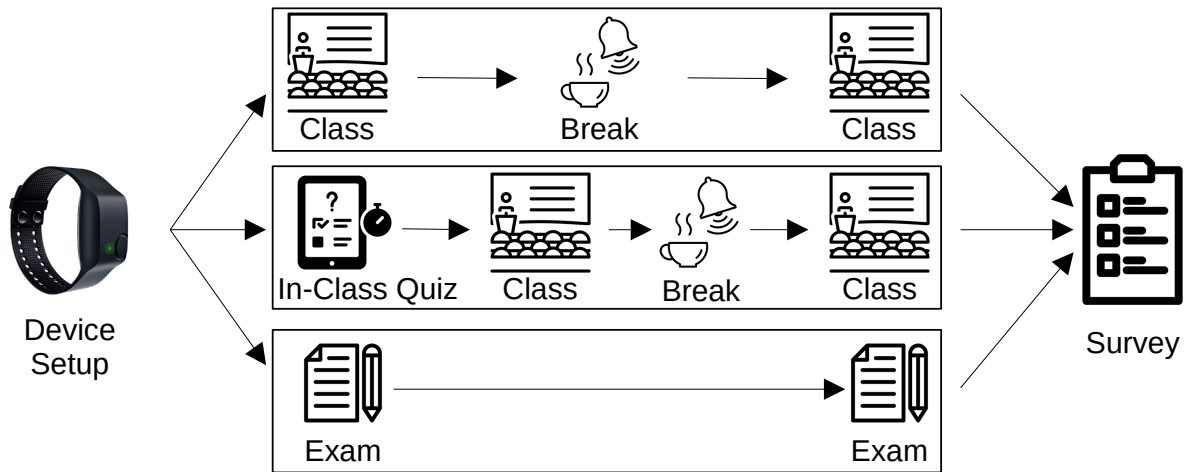
## 1 INTRODUCTION

Modern wearable devices enable the recording of physiological data from users in a continuous and unobtrusive manner. From movement to heart rate and electrodermal activity (EDA), these data can be used to infer users' emotional state, cognitive load, and physical activities. Previous work has also used physiological data to assess participants' engagement during a presentation [2] and the emotional climate of a classroom through physiological synchrony [1]. In this context, using individual wearable devices to monitor students' psycho-physiological states could be helpful for personalised learning interventions. For example, by detecting when students struggle with class content or lose their attention, we could design interventions to improve their understanding and memory recall of the challenging topics.

Working towards personalised interventions for students, we designed a study to measure students' physiological signals, emotions, attention, and engagement in class along with their daily behaviour (e.g., smoking habits, exercise habits, and caffeine and food intake) and class performance (e.g., answers to quizzes, exam grades).

## 2 METHODS

We carried out a longitudinal study throughout a university semester with 44 participants (including two lecturers) in two courses. The courses were at the Bachelor's and Master's levels at the Università



**Figure 1: Data collection procedure.** Each session begins with the device setup. There are three types of sessions: class-only sessions separated by a break, class sessions preceded by a quiz, and exam sessions. A survey followed all sessions.

della Svizzera italiana (USI). We collected data for 26 sessions from each course. The sessions also included the classes' examinations, like in-class quizzes, midterm, and final exams. The data-collection procedure is described by Figure 1. We measured the participants' physiological signals during each session using an E4 wristband from Empatica<sup>1</sup>. The students answered surveys about their habits and personality before and after the study. They also completed the *Big Five* [5] questionnaire. Additionally, students answered surveys about their behaviour (e.g., nutrition, physical activity, sleep quality, and study habits) every day and about their post-class emotional states (e.g., positive and negative affect, engagement, satisfaction) on lecture days. Lecturers also answered after class surveys about their emotions, their engagement, and the engagement they perceived from the students. We recorded audio of the lecturer, video recordings of the classrooms, and a video stream of the slides for each lecture.

Ethics delegates at our Faculty examined and approved the study and its protocol. Students were compensated for their participation. Instructors from the course did not have any access to students' data.

The complete LAUREATE dataset consists of the following items:

- 1400 hours of raw physiological data (heart rate, electrodermal activity, wrist-based accelerometer, and skin temperature) from 44 participants (42 students and 2 lecturers).
- 175 features extracted every 120 seconds with a sliding window of 10 seconds from the physiological data (heart rate variation -HRV-, movement intensity, EDA peak analysis and other statistical descriptors)
- 3600 completed daily surveys from students
- 42 pre-experiment surveys and 35 post-experiment surveys from students
- 32 completed post-session surveys from lecturers
- 70 hours of audiovisual recordings from the classes

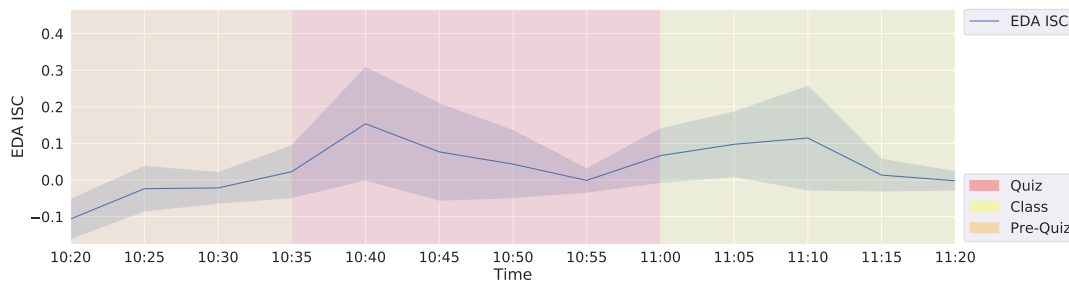
<sup>1</sup>The E4 has the following sensors: Electrodermal Activity, Heart Rate, Skin Temperature, and Accelerometer

- 67 hours of audio recordings of lecturers
- 200 students' answers from 13 in-class quizzes, along with the correct solutions
- Timestamped labels of in-class events (explained topics, breaks, examinations, quiz corrections).
- Students' grades for quizzes, assignments, and midterm and final exams

To the best of our knowledge [7, 8], this is the largest dataset for physiological data from the Empatica E4 device. There are similar multimodal datasets, like CLAS [4], DEAP [3], MMSE [9] or WESAD [6]. Although CLAS and MMSE benefit from having a larger amount of participants, our dataset contains longer in length and repeated sessions per participant. WESAD and DEAP, while comparable on the amount of subjects, were done in a controlled laboratory setting, whereas our work was carried out in a classroom.

### 3 ANALYSIS AND DISCUSSION

Our initial analysis showed that the students' physiological state changed based on the class type (lecture, lab, quiz, exam). For example, we noticed an increment in the sweating rate (a sign of increased stress and arousal) when the students received feedback about a quiz they had just done. This observation is consistent with a focused class paying attention to a lecturer and trying to learn from their recent mistakes. Similarly, students' HRV measurements were lower during exam and quiz segments (indicating more stressful periods) compared to typical class segments. Furthermore, by analysing cross-participant synchrony based on the physiological data, we identified each student's synchronisation level with the overall group and the lecturer. Such analysis may help identify "out-of-sync" moments for each student, which would be useful for developing personalised interventions.



**Figure 2: Higher inter-subject correlation in electrodermal activity during stressful moments. The first peak, visible at 10:40, coincides with the start of a quiz. The second peak, at 11:10, is reached when the Lecturer explains to the students the answers to the previous quiz.**

## 4 CONCLUSIONS

We plan to make the dataset available for scientific purposes under a sharing agreement. Besides the initial intention of the dataset, i.e., to design post-class interventions to improve students' understanding and memory recall, the dataset is quite versatile and enables applications in many domains, including: affective computing (e.g., modelling stress, cognitive load and emotions), privacy (e.g., user identification), modelling behaviour (e.g., exercise level monitoring, sleep and food intake analysis) and learning performance (e.g., grade prediction), context recognition (e.g., class vs exam vs breaks), multimedia signal analysis (e.g., speech and video processing), and the development of multi-modal machine learning algorithms (e.g., combining wearable, audio and video data).

## ACKNOWLEDGMENTS

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